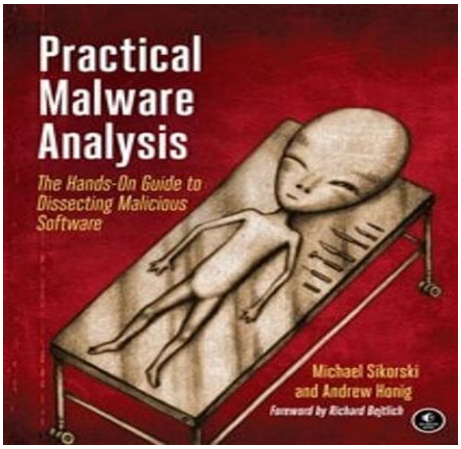
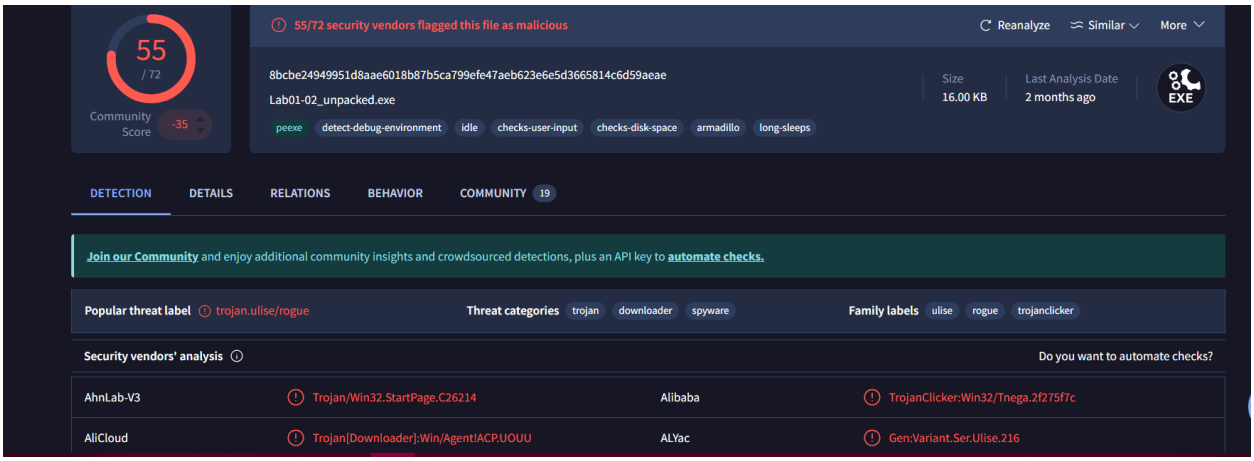
PMA Lab 1.2 Writeup





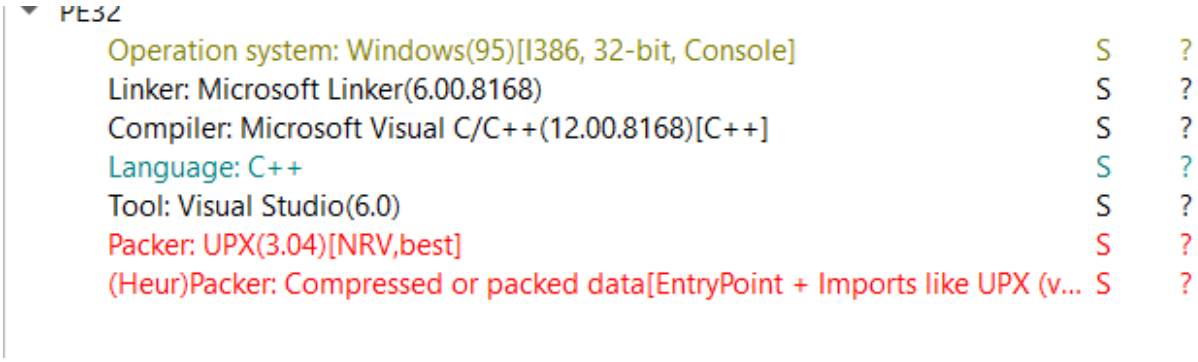
https://www.linkedin.com/in/talha-yaqoob

1. Upload the Lab01-02.exe file to http://www.VirusTotal.com/. Does it match any existing antivirus definitions?



(only shows after being unpacked)

2. Are there any indications that this file is packed or obfuscated? If so, what are these indicators? If the file is packed, unpack it if possible.



Yes, the file is packed by UPX

```
FLARE-VM 01/12/2025 17:59:49.28
C:\Users\MA\Desktop\Practical Malware Analysis Labs\BinaryCollection\Chapter_1L>upx -d Lab01-02.exe
                       Ultimate Packer for eXecutables
                          Copyright (C) 1996 - 2025
UPX 5.0.2       Markus Oberhumer, Laszlo Molnar & John Reiser   Jul 20th 2025

        File size         Ratio      Format      Name
   --------------------   ------   -----------   -----------
     16384 <-      3072   18.75%    win32/pe     Lab01-02.exe

Unpacked 1 file.

FLARE-VM 01/12/2025 18:00:41.14
C:\Users\MA\Desktop\Practical Malware Analysis Labs\BinaryCollection\Chapter_1L>
```

We will use the upx to unpack the file itself

The command is upx –d filename.txt

3. Do any imports hint at this program's functionality? If so, which imports are they and what do they tell you?



imports > flag,CreateServiceA | InternetOpenA | InternetOpenUrlA | OpenSCManagerA | StartServiceCtrlDispatcherA

as we can see the apis

it is registering itself as a service by service apis

OpenSCManagerA and CreateServiceA: These functions are used to communicate with the Service Control Manager (SCM) and register a new executable as a permanent Windows Service. This is the most common technique malware uses to gain persistence (survive a reboot) and ensure its code runs with **SYSTEM-level privileges**

StartServiceCtrlDispatcherA: This is the function called by an application *itself* to actually start running its code as a service, after being installed by CreateServiceA

. and is trying to access a specific url on the internet which is



4. What host- or network-based indicators could be used to identify this malware on infected machines?

CreateServiceA	x	implicit	-	0x0000223C	0x00000000	ADVAPI32.dll
StartServiceCtrlDispatcherA	x	implicit	-	0x0000224C	0x00000000	ADVAPI32.dll
OpenSCManagerA	x	implicit	-	0x0000226A	0x00000000	ADVAPI32.dll

These are the host based indicators they show that:

A new service is being added that access the a specific url on the internet

These functions are part of **advapi32.dll** and are commonly used by legitimate software to run background tasks.

InternetOpenUrlA	x	implicit	-	0x0000232A	0x00000000	WININET.dll
InternetOpenA	x	implicit	-	0x0000233C	0x00000000	WININET.dll

These apis are used to access the internet and open the specific url

string > url-pattern	http://www.malwareanalysisbook.com



Which leads to this